

Assignment - 16 (Day - 16)

Assignment 1: NumPy Array Surgery

The Dataset:

```
np.random.seed(42)
data_array = np.random.randint(1, 501, size=(10, 10))
```

The Question: Using the `data\_array` provided:

1. Find the global mean of the entire array. Replace all elements in the array that are greater than this mean with 0\$.
2. Calculate the sum of each column and the standard deviation of each row .
3. Slice the array to extract the center 4\$ $\times$ 4\$ sub-matrix (rows 3-6, columns 3-6) and convert it into a flat 1D array.

Code:

numpy_operations.py

```
import numpy as np

# Set the random seed and generate the initial array
np.random.seed(42)
data_array = np.random.randint(1, 501, size=(10, 10))

# 1. Find global mean and replace elements greater than the mean with 0
global_mean = data_array.mean()
data_array[data_array > global_mean] = 0

# 2. Calculate sum of each column and standard deviation of each row
col_sums = data_array.sum(axis=0)
row_stds = data_array.std(axis=1)

# 3. Slice center 4x4 sub-matrix (rows 3-6, columns 3-6) and flatten to 1D
# Note: In Python indexing, rows/columns 3-6 correspond to index bounds [3:7]
center_matrix = data_array[3:7, 3:7]
flat_center = center_matrix.flatten()

# Print results
print("Global Mean:", global_mean)
print("Column Sums:", col_sums)
print("Row Standard Deviations:\n", row_stds)
print("Flattened Center Matrix:\n", flat_center)
```

Outputs:

(.venv) PS C:\Wipro Training\Assignments\assignment_16> python numpy_operations.py

Global Mean: 253.7

Column Sums: [557 919 136 524 547 456 947 309 968 530]

Row Standard Deviations:

[61.22752649 75.61375536 69.04172651 98.24810431 85.61401754 77.72522113

82.67744553 30.28283342 61.09304379 88.9]

Flattened Center Matrix:

[0 22 253 236 188 0 0 190 244 0 131 0 0 89 0 14]

Assignment 2: Pandas Data Cleaning & Transformation

The Dataset:

```
raw_data = { 'Product': ['Laptop', 'Mouse', 'Monitor', 'Keyboard', 'Laptop', 'Mouse', None],
              'Revenue': [1200, 25, 300, 75, 1150, None, 150], 'Cost': [800, 10, 200, 40, 850, 15, 100],
              'Date': ['2025-01-10', '2025-02-15', '2025-03-20', '2025-10-05', '2025-11-12', '2025-12 25',
                       '2025-06-01'] }
```

```
df = pd.DataFrame(raw_data)
```

The Question:

1. Handle Missing Values: Fill the missing `Revenue` with the median value of the column and the missing `Product` name with "Unknown".
2. Feature Engineering: Create a column `"Profit"` ($\text{Revenue} - \text{Cost}$) and a column $\text{"Margin_Percentage"} \left(\left(\frac{\text{Profit}}{\text{Revenue}} \right) \times 100\% \right)$.
3. Filtering: Convert the `Date` column to datetime objects. Filter the DataFrame to show only products sold in Q4 (October, November, December) that had a `Profit` greater than \$50\$.

Code:

pandas_operations.py

```
import numpy as np

# Set the random seed and generate the initial array
np.random.seed(42)
data_array = np.random.randint(1, 501, size=(10, 10))

# 1. Find global mean and replace elements greater than the mean with 0
global_mean = data_array.mean()
data_array[data_array > global_mean] = 0

# 2. Calculate sum of each column and standard deviation of each row
```

```

col_sums = data_array.sum(axis=0)
row_stds = data_array.std(axis=1)

# 3. Slice center 4x4 sub-matrix (rows 3-6, columns 3-6) and flatten to 1D
# Note: In Python indexing, rows/columns 3-6 correspond to index bounds [3:7]
center_matrix = data_array[3:7, 3:7]
flat_center = center_matrix.flatten()

# Print results
print("Global Mean:", global_mean)
print("Column Sums:", col_sums)
print("Row Standard Deviations:\n", row_stds)
print("Flattened Center Matrix:\n", flat_center)

```

Outputs:

(.venv) PS C:\Wipro Training\Assignments\assignment_16> python pandas_operations.py
Cleaned DataFrame:

	Product	Revenue	Cost	Date	Profit	Margin_Percentage
0	Laptop	1200.0	800	2025-01-10	400.0	33.333333
1	Mouse	25.0	10	2025-02-15	15.0	60.000000
2	Monitor	300.0	200	2025-03-20	100.0	33.333333
3	Keyboard	75.0	40	2025-10-05	35.0	46.666667
4	Laptop	1150.0	850	2025-11-12	300.0	26.086957
5	Mouse	225.0	15	2025-12-25	210.0	93.333333
6	Unknown	150.0	100	2025-06-01	50.0	33.333333

Filtered Q4 DataFrame:

	Product	Revenue	Cost	Date	Profit	Margin_Percentage
4	Laptop	1150.0	850	2025-11-12	300.0	26.086957
5	Mouse	225.0	15	2025-12-25	210.0	93.333333

Assignment 3: Grouping & Pivot Tables

The Dataset:

```

employee_data = { 'Dept': ['IT', 'HR', 'IT', 'Sales', 'HR', 'IT', 'Sales', 'Sales'],
                  'Status': ['FT', 'FT', 'Contract', 'FT', 'Contract', 'FT', 'Contract', 'FT'],
                  'Salary': [95000, 60000, 70000, 80000, 55000, 98000, 72000, 85000] }
emp_df = pd.DataFrame(employee_data)

```

The Question:

1. Group the data by `Dept` and find the average salary and maximum salary for each department.

2. Create a Pivot Table that shows the count of employees, with `Dept` as the index (rows) and `Status` as the columns.
3. Sort the pivot table to identify which department has the highest number of `Contract` workers.

Code:

pivot_tables.py

```
import pandas as pd

employee_data = {
    'Dept': ['IT', 'HR', 'IT', 'Sales', 'HR', 'IT', 'Sales', 'Sales'],
    'Status': ['FT', 'FT', 'Contract', 'FT', 'Contract', 'FT', 'Contract',
'FT'],
    'Salary': [95000, 60000, 70000, 80000, 55000, 98000, 72000, 85000]
}
emp_df = pd.DataFrame(employee_data)

# 1. Group by Dept and find average (mean) and maximum salary
salary_summary = emp_df.groupby('Dept')['Salary'].agg(['mean', 'max'])

# 2. Pivot Table showing employee count by Dept and Status
pivot_table = emp_df.pivot_table(index='Dept', columns='Status',
aggfunc='size', fill_value=0)

# 3. Sort the pivot table by 'Contract' workers in descending order
sorted_pivot = pivot_table.sort_values(by='Contract', ascending=False)

print("Salary Summary by Dept:\n", salary_summary)
print("\nSorted Pivot Table:\n", sorted_pivot)
```

Outputs:

(.venv) PS C:\Wipro Training\Assignments\assignment_16> python pivot_tables.py

Salary Summary by Dept:

	mean	max
Dept		
HR	57500.000000	60000
IT	87666.666667	98000
Sales	79000.000000	85000

Sorted Pivot Table:

	Status	Contract	FT
Dept			
HR		1	1
IT		1	2
Sales		1	2